

NUnit and Moq

1. NUnit-Handson

Calculator.cs :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace CalcLibrary
{
    public class Calculator
    {
        public int Add(int num1, int num2)
        {
            return num1 + num2;
        }
    }
}
```

CalculatorTests :

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using NUnit.Framework;
using CalcLibrary;

namespace CalcLibrary.Tests
{
    [TestFixture]
    public class CalculatorTests
    {
        private Calculator calculator;

        [SetUp]
        public void Init()
        {
            calculator = new Calculator();
        }

        [TearDown]
        public void Cleanup()
        {
            calculator = null;
        }

        [Test]
```

```

[TestCase(10, 20, 30)]
[TestCase(5, 5, 10)]
[TestCase(100, 200, 300)]
public void Add_ReturnsExpectedResult(
    int num1,
    int num2,
    int expectedResult)
{
    int actualResult =
        calculator.Add(num1, num2);

    Assert.That(
        actualResult,
        Is.EqualTo(expectedResult));
}
}
}

```

Output :

